A product over a finite field

Artin, 13.6.10

Let K be a finite field. Prove that the product of the nonzero elements of K is -1.

Proof. Let

$$P = \prod_{\substack{x \in K \\ x \neq 0}} x$$

be the product of the nonzero elements of K.

We first record a key fact. If $x \in K$, $x \neq 0$, and $x^{-1} = x$, then $x = 1$ or $x = -1$. Indeed, since $x^{-1}x = 1$ and $x^{-1} = x$, we obtain

$$x^2 = x \cdot x = 1.$$

Hence

$$x^2 - 1^2 = 0.$$

Factoring the difference of squares gives

$$(x+1)(x-1) = 0.$$

Since K is a field, one of the factors must be 0. If $x + 1 = 0$, then $x = -1$; if $x - 1 = 0$, then $x = 1$. This proves the claim.

We now distinguish cases according to whether $-1 = 1$ in K.

Case 1: $-1 = 1$**.**

Consider the involution on the nonzero elements of K given by $x \mapsto x^{-1}$. It maps nonzero elements to nonzero elements, it satisfies $(x^{-1})^{-1} = x$, and each pair $\{x, x^{-1}\}$ contributes $x \cdot x^{-1} = 1$ to the product. If $x^{-1} = x$ for some nonzero x, then by the key fact $x = 1$ or $x = -1$; in the present case $-1 = 1$, so in either event $x = 1$, which also contributes 1 to the product. Thus the pairing by inversion shows that

$$P = 1.$$

Since $-1 = 1$ in this case, we conclude that $P = -1$.

Case 2: $-1 \neq 1$**.**

As elements of K, both 1 and -1 are nonzero, and by assumption they are distinct. Let

$$S = \{x \in K : x \neq 0,\ x \neq 1,\ x \neq -1\},$$

and consider

$$P_S = \prod_{x \in S} x.$$

We claim that $P_S = 1$. Again pair each $x \in S$ with x^{-1}. First, x^{-1} is nonzero. Moreover, x^{-1} still lies in S: if $x^{-1} = 1$, then taking inverses gives $x = 1^{-1} = 1$, a contradiction; and if $x^{-1} = -1$, then taking inverses gives $x = (-1)^{-1} = -1$, also a contradiction. The map $x \mapsto x^{-1}$ is an involution since $(x^{-1})^{-1} = x$. It

has no fixed point on S: if $x^{-1} = x$, then by the key fact $x = 1$ or $x = -1$, both of which are excluded from S. Hence S is partitioned into pairs $\{x, x^{-1}\}$ with $x^{-1} \neq x$, and each pair contributes $x \cdot x^{-1} = 1$ to the product. Therefore $P_S = 1$.

Finally, separating first the factor 1 and then the factor -1 from the product of all nonzero elements, we obtain

$$P = 1 \cdot \left((-1) \cdot \prod_{x \in S} x \right) = 1 \cdot \big((-1) \cdot P_S \big).$$

Since $P_S = 1$, this gives

$$P = 1 \cdot ((-1) \cdot 1) = -1.$$

Thus in both cases the product of the nonzero elements of K is -1. $\square$